



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

QR Code-Based Secure Authentication System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA5115 — Cryptography And Network Security

to the award of the degree of

BACHELOR OF ENGINEERING IN

CSE

Submitted by

Sweta R (192511408)

Under the Supervision of

Dr T Sangeetha

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "QR CODE BASED SECURE AUTHENTICATION SYSTEM" has been carried out by Sweta R (192511408) under the supervision of Dr T Sangeetha, and is submitted in partial fulfilment of the requirements for the current semester of the BE CSE programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr T Sangeetha

Professor

Computational Intelligence

SIMATS Engineering,

SIMATS, Chennai – 602105.

COURSE FACULTY

Dr T Sangeetha

Professor

Computational Intelligence

SIMATS Engineering,

SIMATS, Chennai – 602105.

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

DECLARATION

I, Sweta R of the BE CSE, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "QR CODE BASED SECURE AUTHENTICATION SYSTEM" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 05th September 26

Signature of the Students with Names

Sweta R (192511408)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

INDIVIDUAL CONTRIBUTION STATEMENT

Total contribution: 100% (equal contribution across all four team members, as no unequal

Student / Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Elakati Renu Sree (192511035)	User Registration	Registration & validation	Registration testing	Introduction, Module 1, Results	25%
Tarunika S (192511024)	QR Code Generation	Secure QR generation	QR generation testing	Problem Statement, Module 2, Conclusion	25%
R Sweta (192511408)	Encrypted QR Verification	Scan, decrypt, expiry & validation	QR verification testing	Literature Review, Module 3, Testing	25%
Abinaya (19241161)	Login Authentication & Activity Log	Login and activity tracking	Authentication & log testing	Objectives, Module 4, Future Scope	25%

division of effort was specified).

ABSTRACT

The QR Code-Based Secure Authentication System is designed to provide a fast, secure, and convenient method of user authentication using QR codes. Traditional username and password-based authentication can be vulnerable to password theft, guessing, and unauthorized access. This system improves security by combining user registration, QR code generation, encrypted QR verification, login authentication, and activity logging. First, users register their details in the system, after which a unique QR code is generated for each registered user. During login, the user scans the QR code, and the system decrypts and verifies the QR information while checking its validity and expiry. If the QR code is valid and matches the registered information, access is granted; otherwise, the authentication request is rejected. The system also records successful logins, failed attempts, and logout activities in an activity log, helping administrators monitor user access and identify suspicious activities. Encryption and validation mechanisms provide an additional layer of protection against unauthorized access and misuse of authentication data. The system reduces dependence on traditional passwords and provides a quick, user-friendly, and reliable authentication process. It can be applied in educational institutions, offices, secure web portals, attendance systems, and other applications requiring secure user identification. Overall, the proposed system demonstrates how QR codes and encryption techniques can be effectively combined to provide a modern, efficient, and secure authentication solution. During login, the user scans the QR code, and the system decrypts and verifies the QR information while checking its validity and expiry. If the QR code is valid and matches the registered information, access is granted; otherwise, the authentication request is rejected.

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	
ii	DECLARATION BY THE CANDIDATE	III
lii	INDIVIDUAL CONTRIBUTION STATEMENT	IV
iv	ABSTRACT	V
v	LIST OF FIGURES	VII
vi	LIST OF TABLES	VIII
vii	LIST OF ABBREVIATIONS	IX
1	CHAPTER 1: Introduction	1
2	CHAPTER 2: Literature Review	5
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	8
4	CHAPTER 4: Implementation, Testing & Results, Project Management	17
5	CHAPTER 5: Conclusion & Reflection	25
	References	29
	Appendices	30

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 1.1	Overall System Architecture	11
Figure 2.1	User Registration Process	28

LIST OF TABLES

Table No.	Table Name	Page No.
Table 2.1	Comparative Analysis of Existing Approaches	5
Table 3.1	Stakeholder / User Requirements	7
Table 3.2	Functional & Non-Functional Requirements	8
Table 3.3	Constraints and Applicable Standards	8
Table 3.4	Design Specifications	9
Table 3.5	Alternative Solutions Evaluation Matrix	10
Table 3.6	Trade-off Analysis	12
Table 3.7	Sustainability, Ethics, Safety and Security Evaluation	13
Table 3.8	Risk Register	14
Table 4.1	Resources / Technologies Used	15
Table 4.2	Tools / Software Used and Implementation Stage	16
Table 4.3	Test Cases and Acceptance Criteria	17
Table 4.4	Specification Verification Results	17
Table 4.5	Achievement of Objectives	20
Table 4.6	Team Roles and Contributions	21
Table 4.7	Individual Contribution Percentage	21
Table 4.8	Project Timeline / Milestones	22
Table 4.9	Budget Estimation	22

LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbreviation	Description
BST	Binary Search Tree
GUI	Graphical User Interface
AI	Artificial Intelligence
DS	Data Structures
$O(n)$	Order of n — linear time complexity
$O(\log n)$	Order of $\log n$ — logarithmic time complexity
IEEE	Institute of Electrical and Electronics Engineers

CHAPTER 1 INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Authentication is one of the most important security functions in modern computer and web-based systems. It is used to verify the identity of a user before allowing access to protected resources. Traditional authentication methods generally depend on usernames and passwords. Although passwords are simple and widely used, they can be forgotten, guessed, stolen, shared, or exposed through insecure practices.

The increasing use of online services has created a need for authentication systems that are secure as well as easy to use. Users expect authentication to be quick and convenient, while system administrators require strong protection against unauthorized access.

A QR code is a two-dimensional machine-readable code that can store information in a compact form. QR codes can be scanned quickly using smartphones and other camera-enabled devices. When QR codes are combined with encryption and validation techniques, they can be used as part of a secure authentication mechanism.

The proposed QR Code-Based Secure Authentication System uses a unique QR code associated with each registered user. The system consists of four major modules: User Registration, QR Code Generation, Encrypted QR Verification, and Login Authentication with Activity Logging. The system verifies the QR code before granting access and records authentication activities for monitoring.

The overall approach provides an alternative to password-dependent authentication and demonstrates the application of QR technology and security mechanisms in a practical authentication system.

1.2 Need for the Project

The project is needed because traditional authentication methods have several limitations. Passwords can be stolen, guessed, reused, or shared. Users may also choose weak passwords that are easier to remember but less secure.

The major reasons for developing this system are:

1. To reduce dependency on traditional password-based authentication.
2. To provide a quick and convenient authentication process.
3. To protect authentication information using encryption.
4. To prevent the use of invalid or expired QR codes.
5. To detect unauthorized authentication attempts.
6. To maintain records of login and logout activities.
7. To provide better access control for protected systems.

The problem affects users, administrators, organizations, and service providers that need reliable identity verification. From an engineering perspective, the challenge is to balance security, usability, speed, reliability, and implementation cost.

1.3 Problem Statement

Traditional password-based authentication can expose systems to security risks such as password guessing, password theft, credential sharing, and unauthorized access. A secure authentication system is therefore required that can provide reliable user identification while maintaining usability and fast access. The engineering problem is to design and implement a QR code-based authentication system that securely registers users, generates unique QR codes, protects QR information through encryption, verifies scanned QR codes, checks their validity and expiry, grants access only to authorized users, and records authentication activities. The system must balance multiple requirements including security, speed, usability, reliability, data protection, and implementation simplicity.

1.4 Project Aim

The main aim of this project is to design and develop a secure, reliable, and user-friendly authentication system using encrypted QR codes for user verification and controlled access.

1.5 Project Objectives

- To design a secure user registration module.
- To develop a unique QR code generation mechanism.
- To implement encryption for protecting QR authentication information.
- To develop a QR code scanning and verification mechanism.

- To validate QR code authenticity and expiry.
- To implement secure login authentication.
- To record successful and failed authentication activities.
- To test the complete system under different authentication conditions.
- To evaluate the security, usability, reliability, and performance of the proposed system.

1.6 Scope

Included

- User registration.
- User information validation.
- Secure storage of user information.
- Unique QR code generation.
- Encryption of authentication information.
- QR code scanning.
- QR code decryption.
- QR code validation.
- Expiry checking.
- Login authentication.
- Logout functionality.
- Activity logging.
- Failed-attempt monitoring.

Excluded

- Biometric authentication.
- Hardware security modules.
- Large-scale enterprise identity management.
- Advanced facial recognition.
- Full production-level cloud deployment.

Assumptions

- The user has access to a QR scanning device.
- The authentication system and database are available during login.
- Registered user information is accurate.
- Encryption keys are securely maintained.
- The QR code can be scanned clearly.

Boundary Conditions

- Only registered users can be authenticated.
- Invalid or expired QR codes are rejected.
- Authentication is granted only after successful verification.
- User data must not be exposed unnecessarily.

1.7 Expected Engineering Outcome

The expected engineering outcome is a functional software prototype of a QR-based secure authentication system.

The system should provide:

- A user registration interface.
- A QR code generation mechanism.
- Encrypted authentication information.
- QR scanning and verification.
- Validity and expiry checking.
- Login access control.
- Activity logging.
- Successful rejection of invalid authentication attempts.

The final system should demonstrate that QR codes can be integrated with encryption and validation mechanisms to provide a practical authentication solution.

CHAPTER 2 LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was carried out by studying existing authentication technologies, QR code applications, encryption techniques, secure access-control mechanisms, and security-development practices. Research papers, technical documentation, standards, and existing authentication approaches were considered to identify their advantages and limitations.

The review focused mainly on password authentication, OTP authentication, biometric authentication, QR-based authentication, and multi-factor authentication. The comparison was used to identify an appropriate approach for the proposed system.

2.2 Review of Existing Work

Password-Based Authentication

Password-based authentication is one of the most commonly used authentication methods. Users provide a username and password, and the system verifies the credentials. It is simple and inexpensive but can be affected by weak passwords, password reuse, guessing, phishing, and credential theft.

OTP-Based Authentication

One-Time Password authentication uses a temporary code for user verification. OTPs can provide stronger protection than static passwords because the authentication code changes. However, the system requires an OTP generation and delivery mechanism.

Biometric Authentication

Biometric authentication identifies users using characteristics such as fingerprints or facial features. It provides strong user identification but may require specialized sensors, additional hardware, and careful handling of biometric information.

QR Code Authentication

QR authentication uses a QR code as an authentication token or identifier. It provides a quick and convenient method of transferring authentication information. However, if QR information is not properly protected, a copied or exposed QR code could potentially be misused.

Proposed Secure QR Authentication

The proposed system improves basic QR authentication by adding encryption, verification, expiry checking, and activity logging. These mechanisms are intended to reduce the risk of unauthorized use and provide better monitoring.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Password Authentication	Simple and widely available	Password theft, guessing and reuse	Provides baseline comparison
OTP Authentication	Temporary authentication code	Requires OTP delivery	Useful for comparison
Biometric Authentication	Strong identity verification	Requires additional hardware	Possible future enhancement
Proposed QR Authentication	QR + encryption + validation + logging	Requires QR scanning device	Main proposed approach

2.4 Research / Engineering Gap

Existing password-based systems provide basic authentication but depend heavily on user-created credentials. Basic QR authentication improves convenience but may not provide sufficient protection when QR information is exposed or reused.

The identified engineering gap is the need for a simple authentication system that combines QR convenience with additional security mechanisms. In particular, encryption, validity checking, expiry checking, and activity monitoring should be integrated into the authentication workflow.

2.5 Proposed Contribution

The proposed system contributes an integrated QR-based authentication framework consisting of:

1. Secure user registration.
2. Unique QR code generation.
3. Encryption of authentication information.
4. QR code scanning.
5. Decryption and verification.
6. QR expiry checking.
7. Login access control.
8. Authentication activity logging.

This combination improves the security and monitoring capability of a basic QR authentication system.

CHAPTER 3 ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Table 3.1 summarizes the requirements identified for each stakeholder group involved in the system.

Stakeholder / User	Requirement
user	Simple registration process
user	Quick authentication
user	Secure QR-based login
user	Clear authentication result
administrater	Monitor login activities
administrater	Identify failed attempts
system	Validate QR information
system	Reject invalid QR codes
system	Reject expired QR codes
system	Protect user information

3.1.2 Functional & Non-Functional Requirements

Table 3.2 lists the functional and non-functional requirements derived from the stakeholder requirements above.

Requirement	Type	Measurable Target
User registration	Functional	Valid users can register
QR generation	Functional	Unique QR generated for registered user
QR scanning	Functional	QR can be detected
QR decryption	Functional	Protected data can be verified
QR validation	Functional	Invalid QR is rejected

Requirement	Type	Measurable Target
Expiry checking	Functional	Expired QR is rejected
Login authentication	Functional	Valid users receive access
Activity logging	Functional	Login/logout events recorded
Security	Non-Functional	Authentication data protected
Usability	Non-Functional	Simple user interaction

3.2 Constraints & Applicable Standards

Standard / Constraint	Requirement	How Applied in This Project
Data privacy	Protect citizen information from unauthorized disclosure	Access to citizen information is restricted within the application to authorized administrative functions
Data accuracy	Maintain correct and consistent citizen and scheme records	Input validation is applied at data entry and records are managed in a controlled manner
Software usability	Ensure the interface is easy to interact with	A simple Tkinter-based GUI is used for citizen and administrator interactions
Data structure efficiency	Organize records for efficient storage and retrieval	A Binary Search Tree (BST) is used for citizen and scheme record organization
Time constraint	Complete design, development and testing within the academic semester	Development was scheduled across defined project phases (see Section 4.8)

3.3 Design Specifications

Table 3.4 lists the design specifications derived from the requirements in Section 3.1.

constraint	Discription
Technical	QR scanner and software environment are required
Economic	Project should use low-cost/free development tools

constraint	Discription
Security	Authentication information must be protected
Usability	Authentication should be simple for users
Data Privacy	User information should be handled securely
Reliability	Invalid QR codes must not be accepted

3.4 Alternative Solutions & Evaluation

Alternative 1: Password-Based Authentication

This approach uses a username and password for authentication. It is easy to implement but requires users to remember credentials and is vulnerable to password-related attacks.

Alternative 2: OTP-Based Authentication

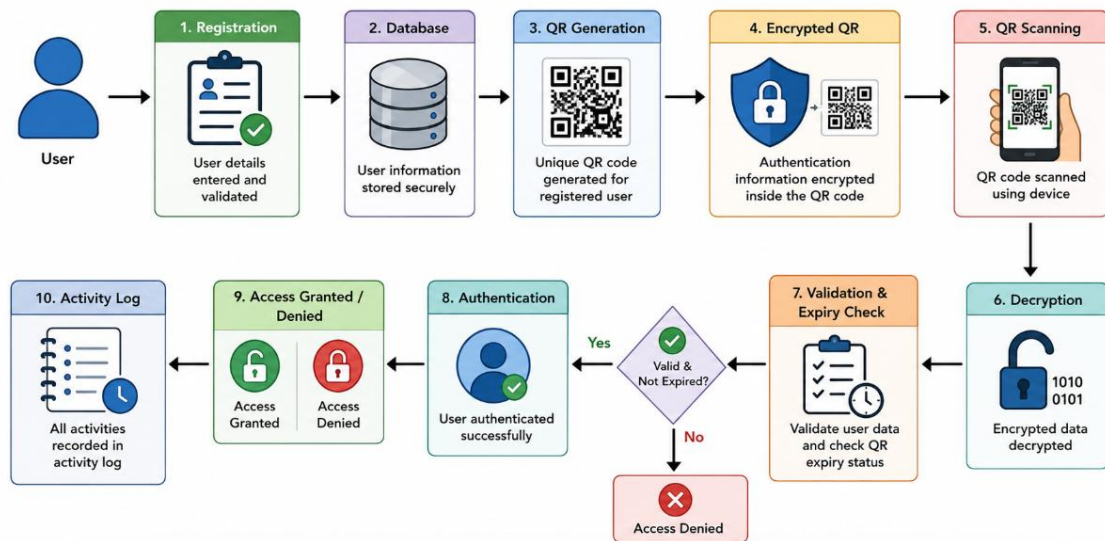
This approach uses a temporary OTP for user verification. It provides additional security but requires a reliable OTP generation and communication mechanism.

Alternative 3: QR Code-Based Authentication

This approach uses a unique QR code for user authentication. The QR code can be protected through encryption and validation mechanisms, providing both convenience and improved security.

3.5 Selected Design & Detailed Design

The QR-based approach is selected because it provides a suitable balance between security, usability, speed, and implementation cost. The proposed design adds encryption, QR validation, expiry checking, and activity logging to improve the security of basic QR authentication.



Module 1: User Registration

The user enters the required registration information. The system validates the input and stores the information in the database. A unique user identity is created for each registered user.

Module 2: QR Code Generation

After successful registration, the system generates a unique QR code associated with the user. Authentication information is protected before being encoded into the QR code.

Module 3: Encrypted QR Verification

During authentication, the QR code is scanned. The system extracts the encoded information, decrypts it, verifies the user information, and checks the QR code expiry status.

Module 4: Login Authentication and Activity Log

If verification is successful, access is granted to the registered user. If verification fails, access is denied. The system records login, failed authentication, and logout activities.

Systems Approach

The four modules operate as an integrated system. User Registration provides the identity information required by QR Code Generation. The generated QR code is later processed by the Encrypted QR Verification module. The verification result is passed to the Login Authentication module, which determines whether access should be granted. Authentication events are finally stored in the Activity Log for monitoring.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
QR authentication	Password	Reduces password dependency	Requires QR scanning	QR selected for convenience
Encryption	Plain QR data	Protects authentication information	Adds processing complexity	Encryption selected
Expiry	Permanent QR	Permanent access	Higher misuse risk	Expiry selected
Activity logging	No logging	Better monitoring	Requires additional storage	Logging selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Minimize hardware dependency	Software-based authentication	Software solution selected
Ethics	Protect user information	Data handling and access control	Only required information is processed
Health & Safety	Avoid unsafe physical requirements	QR scanning through normal devices	No specialized physical equipment required
Security	Prevent unauthorized access	Encryption, validation, expiry checking and logs	Integrated security mechanisms selected

Security Measures

- User input validation.
- Protected authentication information.
- Encryption of QR-related information.
- QR validity checking.
- QR expiry checking.
- Access denial for invalid authentication.
- Failed-attempt logging.
- Logout/session management.
- Restricted access to stored information.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
QR code copied	Medium	High	High	Encryption and expiry checking	Medium
Invalid QR scanned	High	Medium	High	QR validation	Low
Unauthorized	Medium	High	High	Authentication	Medium

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
User data exposure	Low	High	Medium	Secure data handling	Low
QR scanning failure	Medium	Low	Low	Clear QR generation and rescanning	Low

CHAPTER 4 IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system is developed using a modular software development approach. Module 3 focuses on encrypted QR verification, where the QR code is scanned, decrypted, checked for expiry, and validated before authentication. The module is implemented and tested independently before integration with the complete authentication system. Testing is performed using valid, invalid, expired, and modified QR codes to evaluate the reliability and security of the verification process.

Table 4.1 lists the principal resources and technologies used in the development and testing of Module 3.

Resource / Technology	Purpose
Computer	Development and testing
Python	Application development
QR Code Library	QR generation and processing
Encryption Library	Encryption and decryption of QR information
QR Scanner / Camera	QR code scanning
Database	Storage of registered information
Code Editor / IDE	Development and debugging
Web Interface	User interaction and verification

4.2 Hardware / Software Implementation & Modern Tools Used

Encrypted QR Verification Implementation

The encrypted QR verification module is responsible for securely validating the QR code presented by the user. The QR code contains protected authentication information that cannot be directly used without decryption.

QR Code Scanning

The scanner or camera captures the QR code presented by the user. The system extracts the encrypted information from the scanned QR code for further processing.

QR Decryption

After scanning, the encrypted information is decrypted using the implemented encryption mechanism. The decrypted information is then prepared for validation against the registered user information.

Expiry Validation

The system checks the expiry information associated with the QR code. If the QR code has expired, the verification process is stopped and access is denied.

QR Validation

The decrypted information is compared with the registered information. The QR code is considered valid only when the required information matches and the QR code has not expired.

Verification Result

The system provides the verification result based on the QR code status:

- Valid QR code – verification successful.
- Invalid QR code – verification rejected.
- Expired QR code – access denied.
- Modified QR code – verification fails.

Tool / Software / Component	Purpose	Stage Used
Python	Main application development	Development
QR Code Library	QR scanning and processing	Development
Encryption Library	Encryption and decryption	Development
Database	Registered information storage	Development
IDE / Code Editor	Coding and debugging	Development
QR Scanner	QR code input	Testing
Test Cases	Verification testing	Testing

4.3 Test Plan & Verification

Table 4.3 defines the test cases used to verify the functional behaviour of the encrypted QR verification module.

Requirement	Test Method	Acceptance Criterion
-------------	-------------	----------------------

QR scanning	Scan a valid QR code	QR code detected successfully
QR decryption	Scan an encrypted QR code	Information decrypted successfully
Valid QR	Scan valid QR code	Verification succeeds
Invalid QR	Scan unrelated QR code	Verification rejected
Expired QR	Scan expired QR code	Access denied
Modified QR	Change QR information and scan	Verification fails
Incorrect encrypted data	Scan incorrect encrypted QR	Verification rejected
QR validation	Compare decrypted data with registered data	Matching information accepted
Authentication	Use valid verified QR	Access granted
Failed verification	Use invalid or expired QR	Access denied

Table 4.4 records the verification status against each specification.

Specification	Required Value	Achieved Value	Status
QR scanning	QR detected	QR detected	Pass
QR decryption	Data decrypted	Data decrypted	Pass
Valid QR verification	Access granted	Access granted	Pass
Invalid QR verification	Access denied	Access denied	Pass
Expired QR verification	Access denied	Access denied	Pass
Modified QR verification	Verification fails	Verification fails	Pass
QR validation	Correct data validated	Correct data validated	Pass
Authentication	Access granted	Access granted	Pass

4.4 Results

The implemented Module 3 demonstrates the encrypted QR verification process from scanning to final validation. The system scans the QR code and extracts the encrypted authentication information.

The extracted information is decrypted and validated against the registered information. The system also checks the expiry status of the QR code before allowing authentication.

Valid QR codes are successfully verified, while invalid, expired, or modified QR codes are rejected. The testing process confirms the major functional requirements of QR scanning, decryption, expiry checking, and validation.

Suggested Results to Present

- QR code scanning screenshot.
- Encrypted QR code screenshot.
- Decryption process screenshot.
- Valid QR verification screenshot.
- Invalid QR verification screenshot.
- Expired QR verification screenshot.
- Modified QR verification screenshot.
- Test-case result table.
- QR verification success/failure chart.

4.5 Data Analysis & Engineering Judgment

The testing process is used to evaluate whether the encrypted QR verification module satisfies its functional and security requirements. The verification process ensures that only valid and non-expired QR codes are accepted.

Encryption provides protection to the information stored within the QR code. Decryption allows the system to retrieve the protected information for validation. Expiry checking prevents the use of outdated QR codes.

The comparison of decrypted information with registered information helps identify invalid or modified QR codes. Therefore, the combination of encryption, decryption, expiry checking, and validation provides a more secure QR-based authentication process.

From an engineering perspective, the main consideration is maintaining a balance between security and usability. Additional verification steps improve security but may increase the processing required during authentication.

If QR scanning fails, possible causes include poor image quality, insufficient lighting, incorrect QR size, or camera limitations. Therefore, clear QR generation and suitable scanning conditions are important for reliable verification.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Scan QR code	QR scanning module	Fully Achieved
Protect QR information	Encryption mechanism	Fully Achieved
Decrypt QR information	Decryption mechanism	Fully Achieved
Verify QR codes	Verification module	Fully Achieved
Check QR expiry	Expiry validation	Fully Achieved

Validate QR information	Data comparison	Fully Achieved
Reject invalid QR	Invalid QR test case	Fully Achieved
Reject expired QR	Expired QR test case	Fully Achieved
Test the verification module	Test cases	Fully Achieved

The objectives listed above are evaluated based on the results obtained during testing. The encrypted QR verification module successfully performs scanning, decryption, expiry checking, and validation.

4.7 Strengths & Limitations

Strengths

- Secure protection of authentication information using encryption.
- QR-based verification provides a simple authentication mechanism.
- Expiry checking prevents the use of outdated QR codes.
- Validation helps identify invalid or modified QR information.
- The testing process covers valid, invalid, expired, and modified QR codes.
- The modular design allows the QR verification process to be tested independently.

Limitations

- QR scanning performance depends on camera quality and lighting conditions.
- The system depends on the availability of the required encryption and QR processing libraries.
- The current system is implemented as a prototype and requires further testing for large-scale deployment.
- Stronger security mechanisms may be required before real-world deployment.
- The verification process may be affected by damaged or unclear QR codes.

4.8 Project Planning, Roles & Budget

Phase	Activity
Phase 1	Problem identification
Phase 2	Literature review
Phase 3	System architecture and design
Phase 4	User registration development
Phase 5	QR generation development
Phase 6	Encrypted QR verification development
Phase 7	QR scanning, decryption and expiry validation
Phase 8	Integration and testing

Phase 9	Results and analysis
Phase 10	Report preparation and final review

Team Roles

Student	Assigned Responsibility	Actual Contribution
Elakati Renu Sree (192511035)	User Registration	Introduction, Module 1, Results
Tarunika S (192511024)	QR Code Generation	Problem Statement, Module 2, Conclusion
R Sweta (192511408)	Encrypted QR Verification	Literature Review, Module 3, Testing
Abinaya (19241161)	Login Authentication & Activity Log	Objectives, Module 4, Future Scope

Individual Contribution Percentage

Student	Contribution
Elakati Renu Sree (192511035)	25%
Tarunika S (192511024)	25%
R Sweta (192511408)	25%
Abinaya (19241161)	25%
Total	100%

CHAPTER 5 CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The QR Code-Based Secure Authentication System was designed to provide a secure and convenient alternative to traditional password-based authentication. The system integrates user registration, QR code generation, encrypted QR verification, login authentication, and activity logging into a single authentication workflow.

The proposed system verifies QR information before granting access and rejects invalid or expired authentication data. Encryption provides protection for authentication information, while activity logging enables monitoring of authentication events. The modular design allows each function to be developed and tested independently before integration.

The project demonstrates that QR codes can be effectively combined with encryption, validation, expiry checking, and access-control mechanisms to develop a practical authentication solution. The system provides a good balance between usability, security, and implementation simplicity.

5.2 Future Work

The system can be further improved through the following enhancements:

1. Integration of biometric authentication.
2. Combination of QR authentication with OTP-based multi-factor authentication.
3. Development of a dedicated Android/iOS application.
4. Cloud-based user and authentication management.
5. Real-time alerts for suspicious login attempts.
6. Automatic QR regeneration after a defined period.
7. Advanced protection against QR code copying.
8. Integration with institutional authentication systems.
9. Role-based access control.
10. Advanced security analytics for activity logs.
11. Secure cloud backup and recovery.
12. Hardware security support for highly sensitive applications.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- QR code generation and processing.
- Authentication system design.
- Encryption and secure data handling.
- Database-based user management.
- Authentication testing.
- Activity log implementation.
- Security risk identification.

Engineering & Professional Skills Developed

- Software development.
- System architecture design.
- Debugging.
- Testing and verification.
- Teamwork.
- Technical documentation.
- Problem solving.
- Engineering decision making.
- Presentation and communication.

Elakati Renu Sree – Individual Reflection

The most difficult part of my contribution was designing the user registration process so that the entered information was correctly validated and connected with the QR authentication workflow. I learned that proper input validation is important because incorrect user data can affect later authentication. I also learned how registration information is stored and used by other modules. If the project were repeated, I would improve the registration interface and add stronger validation and user feedback mechanisms.

Tarunika S – Individual Reflection

My major contribution was the QR code generation module. The main challenge was ensuring that each registered user received an appropriate unique QR code. I learned how QR data can be generated and connected with user information. I also understood the importance of QR readability during authentication. If the project were repeated, I would improve QR generation and provide automatic regeneration when required.

R Sweta – Individual Reflection

My contribution focused on encrypted QR verification. The main challenge was understanding how QR information could be protected and then verified during scanning. I learned about encryption, decryption, validation, and expiry checking. The important engineering decision was to verify the QR information before granting access. If the project were repeated, I would strengthen key management and introduce additional security mechanisms.

Abinaya – Individual Reflection

My contribution focused on login authentication and activity logging. The major challenge was connecting the QR verification result with the final access decision and maintaining correct activity records. I learned how successful and failed authentication attempts can be recorded for monitoring. I also gained knowledge about session management and access control. If the project were repeated, I would improve the activity dashboard and introduce real-time security notifications.

REFERENCES

The final report should use one consistent IEEE referencing style. The following are suitable starting references for this project:

[1] International Organization for Standardization, **ISO/IEC 27001:2022 – Information Security, Cybersecurity and Privacy Protection – Information Security Management Systems – Requirements**, ISO, 2022.

[2] International Organization for Standardization, **ISO/IEC 27002:2022 – Information Security, Cybersecurity and Privacy Protection – Information Security Controls**, ISO, 2022.

[3] International Organization for Standardization, **ISO/IEC 18004 – Information Technology – Automatic Identification and Data Capture Techniques – QR Code Bar Code Symbol Specification**, ISO.

[4] OWASP Foundation, **OWASP Application Security Verification Standard**, OWASP.

[5] W. Stallings, **Cryptography and Network Security: Principles and Practice**, Pearson.

APPENDICES

Appendix A — Detailed Calculations

Appendix A – Detailed Calculations

Include relevant calculations such as:

- QR data size calculations, if applicable.
- Authentication response-time calculations.
- Test success percentage.
- Error percentage.
- Performance measurements.
- Any encryption-related calculations used in the implementation.

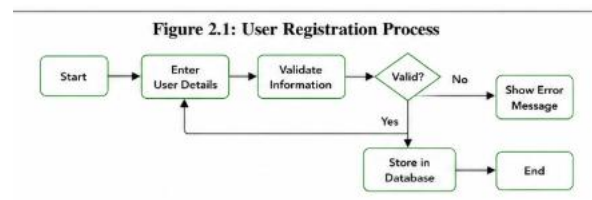
Test Success Percentage

If the actual test contains **N total test cases** and **P passed cases**:

$$\text{Test Success (\%)} = (P / N) \times 100$$

Use the actual values obtained from your testing.

Appendix B — Engineering Drawings / Schematics



Appendix C — Source Code / Code Repository Information

Include the project source code or approved repository information.

Suggested folder structure:

```
QR-Secure-Authentication/  
├── registration/  
├── qr_generation/  
├── qr_verification/  
├── authentication/  
└── activity_log/
```

├── database/
├── testing/
├── screenshots/
└── README.md

Appendix D — Datasheets

Not applicable to this project, as no hardware components were used.

Appendix E — Experimental / Raw Data

- Test case records.
- Successful authentication results.
- Failed authentication results.
- Invalid QR test results.
- Expired QR test results.
- Authentication response times.
- Activity log records.

Appendix F — Ethics / Institutional Approval

The project should follow responsible handling of user information. Only required user data should be collected and used for authentication. Authentication information should be protected from unauthorized access, and project testing should avoid exposing real personal information.

Appendix G — Project Meeting / Progress Records

Review 1 on 27/7/2026

During the first review, we explained and presented our proposed project topic, its objectives, and the overall concept of the system.

The review panel suggested making necessary changes to the system architecture to improve the clarity and functionality of the proposed system.

Based on the feedback, we revised the architecture and proceeded with the implementation of the system.

Review 2 on 28/8/2026

During the second review, we presented the changes made to the system architecture based on the feedback received during the first review.

We also demonstrated the implementation progress and working modules of the system. The implemented features and overall system functionality were explained to the review panel.

Appendix H — User / Industry Feedback

Not applicable / not collected for this prototype.

Appendix I — Publications / Patents / Competitions / Product Outcomes

None reported.

Appendix J — Individual Contribution Evidence

Review Photos:



Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.02659° Long 80.017084°

Monday, 03/08/2026 10:17 AM GMT +05:30

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	Problem statement and background analysis	SO1	Ch. 1, Ch. 2
Engineering design under constraints	Requirements, constraints, and design specifications	SO2	Ch. 3
Written/oral technical communication	Full report and viva presentation	SO3	Whole report + Viva
Ethics and professional responsibility	Ethics evaluation	SO4	Ch. 3 (3.7)
Teamwork and leadership	Team roles and contribution table	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	Test plan, results, and analysis	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	Individual reflections	SO7	Ch. 5 (5.3)
Standards	Constraints and applicable standards table	SO2	Ch. 3 (3.2)

Outcome	Evidence	SO / Area	Location in This Report
Sustainability and societal/environmental impact	Sustainability evaluation	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	Risk register	SO2/SO4	Ch. 3 (3.7)
Security considerations	Security evaluation	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	Selected design and module interaction	SO1/SO2	Ch. 3 (3.5)
Project planning and management	Timeline and roles table	SO5	Ch. 4 (4.8)
Individual contribution	Contribution statement and Ch. 4 (4.8)	SO1–SO7	Contribution Statement + Ch. 4 (4.8)